would pry the planet and its sun apart, sending them onward in markedly different orbits. Therefore, like Miller's planet, it must be heated and lit by Gargantua's anemic accretion disk.

The Trip to Mann's Planet

The *Endurance*'s trip to Mann's planet begins near Gargantua and ends far from it. Such a trip—in my scientist's interpretation of the movie—requires two gravitational slingshots (Chapter 7), one at the beginning of the trip and one at the end.

At the beginning, the challenges are twofold: In its parking orbit near Gargantua, the *Endurance* is moving at a third of light speed, $c/3$, in the wrong direction, a circumferential orbit around Gargantua; it must be deflected into radial motion, away from Gargantua. And the *Endurance* isn't moving fast enough. Gargantua's gravitational pull is so strong that, if the *Endurance* is deflected onto a radial trajectory but still has its starting speed of $c/3$, then Gargantua will pull it to a halt by the time it has traversed only a small fraction of the distance to Mann's planet. To overcome Gargantua's gravity and reach Mann's planet moving with the same speed as the planet, roughly $c/20$, the first slingshot must accelerate the *Endurance* up to nearly half the speed of light. To achieve this, Cooper must find an intermediate-mass black hole (IMBH) at an appropriate location and moving with a suitable velocity.

Finding the necessary IMBH is not easy, and having found it, reaching it at the right point and moment in its orbit may not be easy. Most of the months'-long trip may be spent reaching the IMBH, and it might entail considerable waiting for the IMBH to arrive. Once the slingshot is completed, the trip to Mann's planet, with speed about $c/2$ initially and gradual slowing to roughly $c/20$, will take roughly an additional forty days.